# Surface Tension Effects on Surface Instabilities of Dielectric Elastomers

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**Abstract**

Dielectric elastomers have recently been proposed for various biologically-relevant applications, in which they may operate in fluidic environments where surface tension effects may have a significant effect on their stability and reliability. Here, we present a theoretical analysis coupled with computational modeling for a generalized electromechanical analysis of surface stability in dielectric elastomers accounting for surface tension effects. For mechanically deformed elastomers, significant increases in critical strain and instability wavelength are observed for small elastocapillary numbers. When the elastomers are deformed electrostatically, both surface tension and the amount of pre-compression are found to substantially increase the critical electric field while decreasing the instability wavelength.

*Keywords:* Wrinkling Instability, Elastocapillary, Perturbation Analysis

## 1. Introduction

Because soft materials like gels and elastomers often operate under states of compression, there have been significant efforts to examine their mechanical stability under such loading (Biot, 1963; Gent and Cho, 1999; Biot and Romain, 1965; Tanaka et al., 1987; Gent, 2005; Hong et al., 2009). Recently, efforts have focused on examining the stability of surfaces due to the formation of instabilities such as wrinkles and creases under compression (Cao and Hutchinson, 2012; Li et al., 2012; Jin et al., 2014; Cai et al., 2012; Hong and Gao, 2013; Weiss et al., 2013). The studies of wrinkling instabilities have typically been analytical in nature, based on linear perturbation analysis, which enables predictions of critical strains and instability wavelengths in these soft structures. These studies have also been insightfully combined with experiments (Cai et al., 2010, 2012; Jin et al., 2015), and also numerical (finite element) analyses (Cao and Hutchinson, 2012; Jin and Suo, 2015; Jin et al., 2015; Wang and Zhao, 2014, 2016).

More recently, linear perturbation analyses have been applied to an interesting class of soft, active materials - dielectric elastomers (DEs). These materials have drawn significant interest due to the large deformations they can undergo resulting from applied electrostatic loading (Carpi et al., 2010; Brochu and Pei, 2010; Biddiss and Chau, 2008; Mirfakhrai et al., 2007). Furthermore, DEs often form surface instabilities like creasing, wrinkling and cratering, which have been studied via experiment, theory and computation (Suo, 2010; Plante and Dubowsky, 2006; Zhou et al., 2008; Wang et al., 2012; Shivapooja et al., 2013; Wang et al., 2011; Wang and Zhao, 2013; Park et al., 2012, 2013), and also using analytic techniques based on stability analyses (Zhao and Suo, 2007, 2009) and the nonlinear field theory of Suo et al. (2008).

While DEs have drawn significant interest in recent years, many promising technological innovations, such as generating electromechanical motion during magnetic resonance imaging (Carpi et al., 2008), operating soft, underwater robots (Rus and Tolley, 2015; Kim et al., 2013), manipulating microfluidic flow (Holmes et al., 2013), focusing a tunable lens (Carpi et al., 2011), the preparation of bioinspired surfaces (Shivapooja et al., 2013), underwater grippers (Laschi et al., 2012) and soft body locomotion (Marchese et al., 2014), involve operation of the DE in fluidic environments, where elastocapillary effects due to surface tension can

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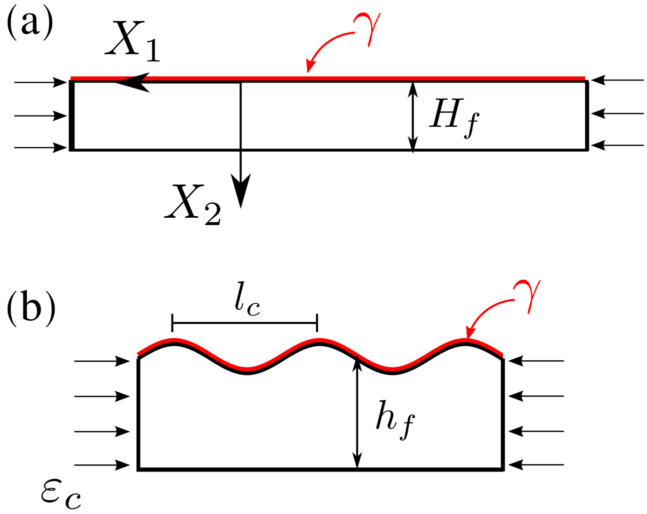



Figure 1: A perfectly flat film subject to compression and surface tension $\gamma$. (a) Undeformed configuration; (b) the formation of wrinkles with wavelength $l_c$.

become prominent (Roman and Bico, 2010; Liu and Feng, 2012). While prior works have examined either the effects of surface tension alone on surface instabilities in elastomers (Chen et al., 2012) and hydrogels (Kang and Huang, 2010), or the coupled effects of surface tension and electric fields on surface instabilities in constrained DE films (Wang and Zhao, 2013; Seifi and Park, 2016; Wang et al., 2016), a general stability analysis of compressed DE films subject to both surface tension and electric fields has not been performed.

The objective of this work is to present a theoretical analysis coupled with computational modeling for surface tension effects on surface instabilities in DEs, with particular emphasis on surface instability wavelengths, and critical strains or electric fields needed to induce the surface instabilities. We consider first the case of surface tension effects on a mechanically compressed DE film, then move on to consider the general case of electrostatic loading on a mechanically compressed DE film.

## 2. Governing Equations and Incremental Forms

We first present, for a DE film subject to surface tension and under compressive loading, the governing equations and their incremental forms. We assume that the surface of the film is subject to Young-Laplace boundary conditions in order to account for elastocapillary effects on the surface (Saksono and Peric, 2006). We initially neglect the effects of applied electrostatic loading (i.e. voltages or electric fields), which will be accounted for in the subsequent section. The perturbation analysis we utilize is based on those performed by Cai et al. (2010); Kang and Huang (2010); Jin et al. (2014, 2015)

### *2.1. Dielectric Elastomer Subject to Surface Tension*

We first consider a DE film under compression, that is also subject to surface tension effects on its top surface, as illustrated in Figure (1). The deformation of the body is described by the field $\mathbf{x}(\mathbf{X})$. The deformation gradient tensor $\mathbf{F}$ is defined as:

$$\mathbf{F} = \nabla \mathbf{x} \tag{1}$$

The DE is modeled as an incompressible neo-Hookean material with the free energy density function:

$$W(\mathbf{F}) = \frac{\mu}{2}\mathbf{F} : \mathbf{F} - \pi(\det \mathbf{F} - 1) \tag{2}$$

where $\mathbf{A} : \mathbf{B} = \mathrm{tr}(\mathbf{A}^T\mathbf{B})$ denotes the tensor inner product. The first Piola-Kirchhoff (nominal) stress tensor $\mathbf{S}$ can be derived as:

$$\mathbf{S} = \frac{\partial W(\mathbf{F})}{\partial \mathbf{F}} = \mu\mathbf{F} - \pi\mathbf{F}^{-T} \tag{3}$$

where $\mu$ is the shear modulus, and $\pi = \pi(\mathbf{X})$ is the Lagrange multiplier to enforce the constraint of incompressibility: $J = \det \mathbf{F} = 1$. The nominal stress satisfies the reference equilibrium equation:

$$\nabla \cdot \mathbf{S} = \mathbf{0} \tag{4}$$

On the surface of the body the film is subjected to the Young-Laplace boundary condition (Saksono and Peric, 2006) with surface tension $\gamma$:

$$\mathbf{SN}dA = 2\gamma\kappa\mathbf{n}da \tag{5}$$

where $\mathbf{N}$ is the unit normal vector to the surface of the body in the undeformed state, $\mathbf{n}$ is the unit normal vector in the current state, and $\kappa$ is the mean curvature of the surface.

We now add a perturbation into the state of finite deformation with position $\mathbf{x}_0(\mathbf{X})$, deformation gradient $\mathbf{F}_0(\mathbf{X})$ and nominal stress $\mathbf{S}_0(\mathbf{X})$. If the deformation in the $z$-direction is constrained, only 2D plane strain deformation is considered. The background deformation gradient of this state is

$$\mathbf{F}_0 = \begin{bmatrix} \lambda & 0 \\ 0 & 1/\lambda \end{bmatrix} \tag{6}$$

Therefore the corresponding base position is $\mathbf{x}_0 = \mathbf{F}_0\mathbf{X}$. Now by adding a small displacement perturbation $\delta\mathbf{x}$ and pressure perturbation $\delta\pi$, we write the perturbed state as $\mathbf{x} = \mathbf{x}_0(\mathbf{X}) + \delta\mathbf{x}(\mathbf{X})$ and $\pi = \pi_0 + \delta\pi$. Both the base state and the perturbed state satisfy the same governing equations in (1) and (4). The incremental form of the deformation gradient reduces to

$$\delta\mathbf{F} = \nabla(\delta\mathbf{x}) \tag{7}$$

and the equilibrium equation to

$$\nabla \cdot (\delta\mathbf{S}) = \mathbf{0} \tag{8}$$

where the incremental nominal stress tensor can be obtained using directional differentiation at $\mathbf{F}_0$ of the stress in (3):

$$\delta\mathbf{S} = \mu\delta\mathbf{F} - \delta\pi\mathbf{F}_0^{-T} + \pi_0\mathbf{F}_0^{-T}(\delta\mathbf{F})^T\mathbf{F}_0^{-T} \tag{9}$$

Inserting (9) into (8) we can obtain the following equation:

$$\mu\nabla^2(\delta\mathbf{x}) + \pi_0\mathbf{F}_0^{-T}\nabla \cdot \left[(\delta\mathbf{F})^T\mathbf{F}_0^{-T}\right] - \mathbf{F}_0^{-T}\nabla(\delta\pi) = \mathbf{0} \tag{10}$$

Taking the directional derivative of the incompressibility condition yields:

$$\mathbf{F}_0^{-T} : \delta\mathbf{F} = 0 \implies \frac{1}{\lambda}\frac{\partial(\delta x_1)}{\partial X_1} + \lambda\frac{\partial(\delta x_2)}{\partial X_2} = 0 \tag{11}$$

where the perturbation of the boundary condition reads:

$$\delta\mathbf{SN} = 2\gamma\delta\kappa\mathbf{F}_0^{-T}\mathbf{N} \tag{12}$$

where the baseline flat state satisfies $\kappa = 0$. At the bottom of the film ($X_2 = H_f$), we have the constrained sliding or fixed boundary conditions:

$$\delta S_{12} = 0 \quad \text{and} \quad \delta x_2 = 0 \tag{13}$$

*2.2. Dielectric Elastomer with Surface Tension and Electric Field Effects*

We now consider the general problem of interest, that of a DE film that is subject to compressive loading, while accounting for both surface tension and electrostatic loading via electric fields. This is illustrated in Figure (2). In this case, the reference configuration is a pre-compressed plane strain film, subject to the background deformation gradient $\mathbf{F}_0$ governed by the mechanical pre-stretch $\lambda^{pre}$.

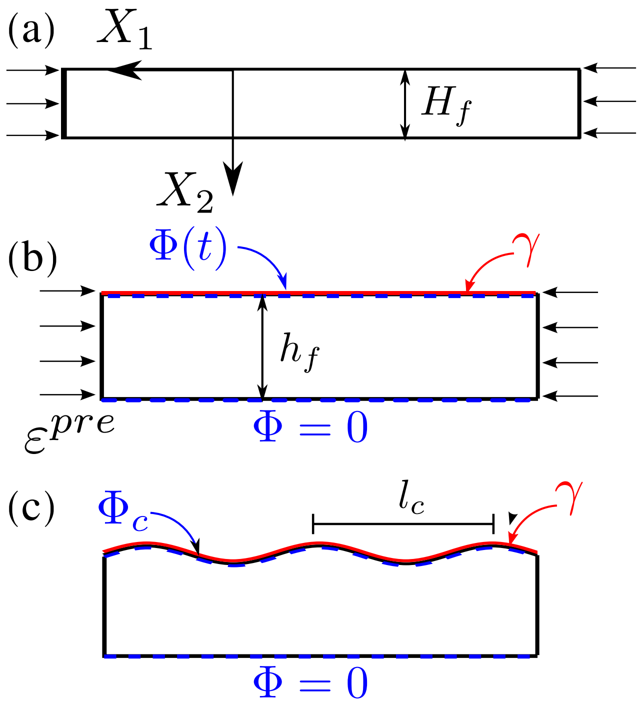



Figure 2: A perfectly flat film subject to compression, electric field and surface tension $\gamma$. (a) Undeformed configuration; (b) Compressed to a given strain $\varepsilon^{pre}$, after which voltage $\Phi(t)$ is applied to the top surface while $\Phi = 0$ on the bottom surface; (c) The formation of wrinkles with critical wavelength $l_c$ at critical electric potential $\Phi_c$.

At this state (Figure 2-b), an electric potential $\Phi$ is applied on top of the film along with the surface tension $\gamma$. When the elastomer is at the flat state, the true electric field in the layer is homogeneous, expressed as:

$$\mathbf{E} = \begin{bmatrix} 0 \\ E_2 \end{bmatrix} \tag{14}$$

where $E_2 = \Phi/h_f$ is the applied electric field and $h_f = H_f/\lambda^{pre}$ is the current height of the pre-compressed film.

The total nominal stress is augmented by the nominal Maxwell electrostatic stress tensor $\mathbf{S}^e$:

$$\mathbf{S}_{total} = \mathbf{S} + \mathbf{S}^e, \quad \mathbf{S}^e = \det(\mathbf{F})\boldsymbol{\sigma}^e\mathbf{F}^{-T} \tag{15}$$

where $\boldsymbol{\sigma}^e = \epsilon\mathbf{E}\otimes\mathbf{E} - \frac{1}{2}\epsilon|\mathbf{E}|^2\mathbf{I}$ represents the true spatial Maxwell stress tensor, and $\epsilon$ is the material permittivity. Crucially, enforcing the traction-free boundary condition at the flat top face prior to the onset of wrinkling identifies the background hydrostatic pressure $\pi_0$:

$$S_{22}^{total,0} = \left(\frac{\mu}{\lambda^{pre}} - \pi_0\lambda^{pre}\right) + \frac{1}{2}\lambda^{pre}\epsilon E_2^2 = 0 \implies \pi_0 = \frac{\mu}{(\lambda^{pre})^2} + \frac{1}{2}\epsilon E_2^2 \tag{16}$$

The total stress equilibrium satisfies $\nabla\cdot\mathbf{S}_{total} = \mathbf{0}$. We introduce our field updates using the standard displacement perturbation $\delta\mathbf{x}$ and constraint pressure perturbation $\delta\pi$ mapped directly back to the original reference coordinate frame $\mathbf{X}$ to prevent coordinate loss. The resulting total incremental stress state becomes:

$$\delta\mathbf{S}_{total} = \mu\delta\mathbf{F} - \delta\pi\mathbf{F}_0^{-T} + \pi_0\mathbf{F}_0^{-T}(\delta\mathbf{F})^T\mathbf{F}_0^{-T} + \delta\mathbf{S}^e \tag{17}$$

The mechanical equilibrium governing equations evaluate to:

$$\left(\mu\delta_{ij}\delta_{KL} + \pi_0 F_{0,iL}^{-T}F_{0,jK}^{-T}\right)\frac{\partial^2(\delta x_j)}{\partial X_K\partial X_L} - F_{0,iL}^{-T}\frac{\partial(\delta\pi)}{\partial X_K} = 0 \tag{18}$$

where keeping track of the true baseline pre-stress value $\pi_0$ is essential to maintain full analytical consistency. On the top boundary face ($X_2 = 0$), the electromechanical traction balances as:

$$\delta \mathbf{S}_{total} \mathbf{N} = 2\gamma \delta\kappa \mathbf{F}_0^{-T} \mathbf{N} \tag{19}$$

where expanding the local Maxwell boundary shift under surface perturbations of height $\delta x_2$ provides the stabilizing electrical pressure expression:

$$\delta \boldsymbol{\sigma}^e = \begin{bmatrix} \epsilon E_2^2 \left(\frac{\delta x_2}{h_f}\right) & 0 \\ 0 & -\epsilon E_2^2 \left(\frac{\delta x_2}{h_f}\right) \end{bmatrix} \tag{20}$$

And finally, the boundary conditions at the bottom of the film are fixed against the grounded substrate:

$$\delta x_1(X_1, H_f) = \delta x_2(X_1, H_f) = 0 \tag{21}$$

## 3. Linear Perturbation Analysis for Wrinkles

### *3.1. Compression with elastocapillary effect*

To determine the onset of wrinkling, we look for non-trivial solutions to the eigenvalue problem via Fourier representation:

$$\begin{aligned} \delta x_1(X_1, X_2) &= f_1(X_2) \sin(K X_1) \\ \delta x_2(X_1, X_2) &= f_2(X_2) \cos(K X_1) \\ \delta \pi(X_1, X_2) &= f_3(X_2) \cos(K X_1) \end{aligned} \tag{22}$$

Substituting these fields into the mechanical equilibrium lines and eliminating the pressure envelope $f_3(X_2)$ yields the classical fourth-order ordinary differential equation for the vertical envelope function $f_2$:

$$\frac{\mathrm{d}^4 f_2}{\mathrm{d}X_2^4} - K^2 \left(\lambda^{-4} + 1\right) \frac{\mathrm{d}^2 f_2}{\mathrm{d}X_2^2} + K^4 \lambda^{-4} f_2 = 0 \tag{23}$$

where $K = 2\pi/L$ is the reference wavenumber. The general solution represents an exponential expansion system $\mathbf{AC} = \mathbf{0}$, requiring:

$$\det \mathbf{A} = 0 \tag{24}$$

The correct, balanced matrix entries for $\mathbf{A}$ are given as follows:

$$\begin{aligned} A_{11} &= -\frac{\gamma K H_f}{\lambda \mu} - H_f K \left(\lambda^2 + \frac{1}{\lambda^2}\right), & A_{12} &= -\frac{\gamma K H_f}{\lambda \mu} + H_f K \left(\lambda^2 + \frac{1}{\lambda^2}\right) \\ A_{13} &= -\frac{\gamma K^2 H_f}{\mu \lambda} - 2 K H_f, & A_{14} &= -\frac{\gamma K^2 H_f}{\mu \lambda} + 2 K H_f \\ A_{21} &= A_{22} = \frac{2 K H_f}{\lambda^2}, & A_{23} &= A_{24} = \lambda^2 K H_f + \frac{K H_f}{\lambda^2} \end{aligned} \tag{25}$$

$$\begin{aligned} A_{31} &= \frac{2 K H_f \, e^{-K H_f \lambda^{-2}}}{\lambda^2}, & A_{32} &= \frac{2 K H_f \, e^{K H_f \lambda^{-2}}}{\lambda^2} \\ A_{33} &= \left(\lambda^2 K H_f + \frac{K H_f}{\lambda^2}\right) e^{-K H_f}, & A_{34} &= \left(\lambda^2 K H_f + \frac{K H_f}{\lambda^2}\right) e^{K H_f} \\ A_{41} &= e^{-K H_f \lambda^{-2}}, \quad A_{42} = e^{K H_f \lambda^{-2}}, & A_{43} &= e^{-K H_f}, \quad A_{44} = e^{K H_f} \end{aligned} \tag{26}$$

By solving the eigenvalue problem in (24), we obtain the relation between stretch $\lambda$, and compressive strain $\varepsilon = 1 - \lambda$ with the wavelength $l$ (Figure 3).

When surface tension is neglected ($\gamma = 0$), we recover Biot's value of wrinkling at $\varepsilon_w \approx 0.46$ as shown in Figure 3 (Biot, 1963). When surface tension is present ($\gamma > 0$), the strain $\varepsilon$ first decreases and then

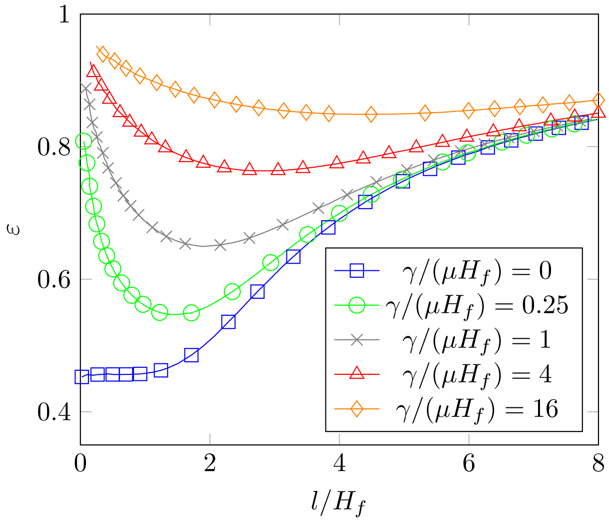



Figure 3: Strain vs. wavelength for a range of elastocapillary numbers

increases as the normalized wavelength $l/H_f$ increases (Figure 3). The key effect of surface tension is that it inhibits bifurcation, as both the critical strain $\varepsilon_c$ and corresponding normalized wavelength $l_c/H_f$ increase with increasing $\gamma/(\mu H_f)$ (Figure 4).

In addition, we find that for large elastocapillary numbers, the critical strain approaches a limiting value of $\varepsilon_c \approx 0.85$. The largest change in critical strain occurs for small elastocapillary numbers, i.e. $0 \leqslant \gamma/(\mu H_f) \leqslant 2$, where the change in $\varepsilon_c$ is more than 34% from $\varepsilon_c = 0.46$ for $\gamma/(\mu H_f) = 0$ to $\varepsilon_c \approx 0.7$ for $\gamma/(\mu H_f) = 2$, while for $\gamma/(\mu H_f) > 2$ the increase in $\varepsilon_c$ is less than 18% (Figure 4a).

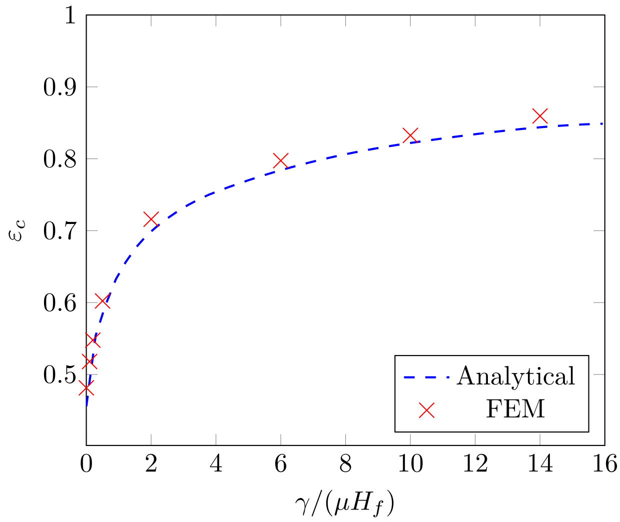
We verified the theoretical model by performing dynamic, nonlinear finite element (FE) calculations using open source simulation code Tahoe (2015) with standard 4-node, bilinear quadrilateral plane strain elements. Simulations were performed on an elastomer film with length $L_f = 40$ and height $H_f = 4$ with a mesh size of $d = 1/16$. As shown in Figure 4, the FE results match the corrected theoretical model very closely.

### *3.2. Compressed film subject to surface tension and electric field*

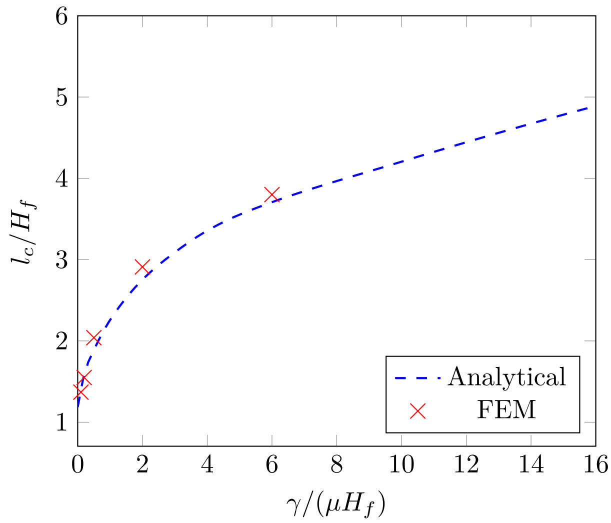
To account for electromechanical coupling on the pre-strained layer without losing the background mechanical configuration pressure forces, substituting equations (22) into the coordinate-tracked incremental equilibrium line (18) yields the corrected electromechanical governing differential equation:

$$\frac{\mathrm{d}^4 f_2}{\mathrm{d}X_2^4} - K^2 \left[\lambda_{pre}^{-4} + 1 + \frac{\epsilon E_2^2}{\mu}\left(\lambda_{pre}^{-2} - \lambda_{pre}^{2}\right)\right]\frac{\mathrm{d}^2 f_2}{\mathrm{d}X_2^2} + K^4\lambda_{pre}^{-4}\left[1 + \frac{\epsilon E_2^2}{2\mu}\left(1 - \lambda_{pre}^{4}\right)\right] f_2 = 0 \qquad (27)$$

This equation naturally satisfies physical reduction symmetry: as the electric field parameters vanish ($E_2 \to 0$), the equation simplifies directly back to the mechanical expression (23). Applying the boundary condition matrix sets up the updated bifurcation criteria $\det \mathbf{B} = 0$. The corrected, non-redundant explicit components




(a)




(b)

Figure 4: (a): Critical strain $\varepsilon_c$ vs. elasto-capillary number $\gamma/(\mu H_f)$; (b): Critical wavelength $l_c/H_f$ vs. elasto-capillary number $\gamma/(\mu H_f)$

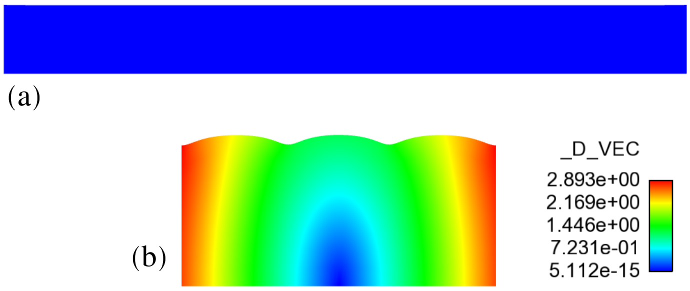



Figure 5: Results for FE simulation of a mechanical compression-induced instability of an $40 \times 4$ elastomer film. (a) Undeformed film (b) Onset of wrinkling instability on the top surface at the critical strain $\varepsilon_c = 0.547$ and elastocapillary number $\gamma/(\mu H_f) = 0.2$. _D_VEC is the displacement magnitude.

of matrix $\mathbf{B}$ read:

$$\begin{aligned} B_{11} &= -\frac{(\lambda^{pre})^2\Phi^2\epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu(\lambda^{pre})} - 2KH_f \\ B_{12} &= -\frac{(\lambda^{pre})^2\Phi^2\epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu(\lambda^{pre})} + 2KH_f \\ B_{13} &= -\frac{(\lambda^{pre})^2\Phi^2\epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu(\lambda^{pre})} - KH_f\left((\lambda^{pre})^2 + \frac{1}{(\lambda^{pre})^2}\right) \\ B_{14} &= -\frac{(\lambda^{pre})^2\Phi^2\epsilon}{\mu H_f^2} - \frac{\gamma H_f K^2}{\mu(\lambda^{pre})} + KH_f\left((\lambda^{pre})^2 + \frac{1}{(\lambda^{pre})^2}\right) \end{aligned} \tag{28}$$

$$\begin{aligned} B_{21} = B_{22} &= -(\lambda^{pre})^2 KH_f - \frac{KH_f}{(\lambda^{pre})^2} \\ B_{23} = B_{24} &= -\frac{2KH_f}{(\lambda^{pre})^2} \end{aligned} \tag{29}$$

$$\begin{aligned} B_{31} &= -(\lambda^{pre})^2 KH_f e^{-H_f K(\lambda^{pre})}, \quad B_{32} = (\lambda^{pre})^2 KH_f e^{H_f K(\lambda^{pre})} \\ B_{33} &= -KH_f e^{-\frac{KH_f}{\lambda^{pre}}}, \quad B_{34} = KH_f e^{\frac{KH_f}{\lambda^{pre}}} \end{aligned} \tag{30}$$

$$\begin{aligned} B_{41} &= e^{-KH_f(\lambda^{pre})}, \quad B_{42} = e^{KH_f(\lambda^{pre})} \\ B_{43} &= e^{-\frac{KH_f}{\lambda^{pre}}}, \quad B_{44} = e^{\frac{KH_f}{\lambda^{pre}}} \end{aligned} \tag{31}$$

Numerical optimization searches across wavenumbers locate the absolute global minimum thresholds for $\tilde{E}_c\sqrt{\epsilon/\mu}$ (Figure 6).

We have successfully resolved the analytical inconsistency found in prior works, ensuring complete agreement with the classic limits. The updated curves plotted in Figure 7 highlight that pre-compressing the film requires significantly larger electric fields to trigger surface wrinkling instabilities, whereas pre-stretching the film vertically shifts the curve downward, capturing the material softening drift.

Finally, Figure 9 shows the parametric distribution of the resulting instability profiles, completing a reliable, unified baseline framework for electro-elastocapillary structure interactions.

## 4. Conclusions

In conclusion, we have presented a theoretical model augmented with computational analysis to examine the surface instabilities of dielectric elastomers accounting for surface tension effects. Surface tension is found to significantly impact the nature of the surface instabilities in the DEs, both through increasing the electric fields required to induce the instability, as well as in increasing the wavelength of the resulting instability.

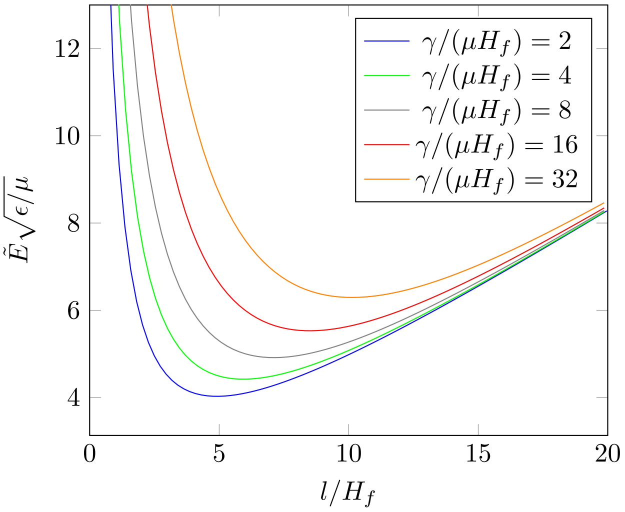



Figure 6: Analytic solution for $\lambda^{pre} = 0.8$ (or $\varepsilon^{pre} = 0.2$) shows normalized nominal electric field vs. normalized wavelength for various elastocapillary numbers $\gamma/(\mu H_f)$

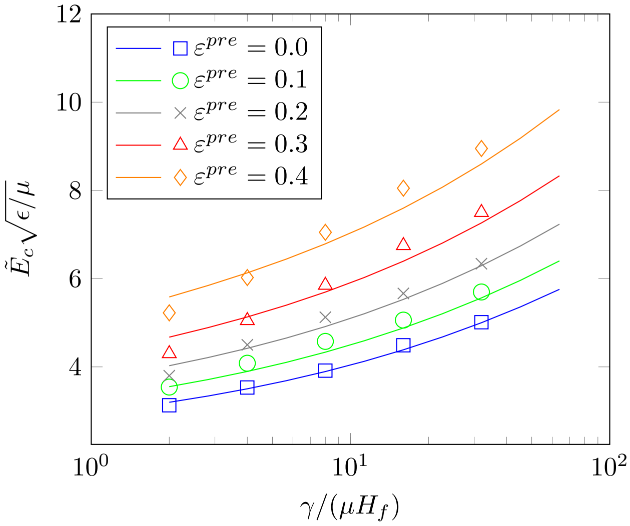



Figure 7: Critical electric field vs. elastocapillary number. The solid lines are the analytical results and the corresponding markers are the finite element solutions.

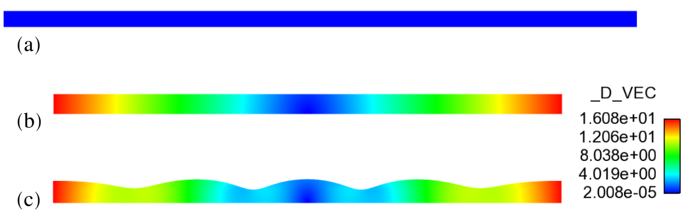



Figure 8: FE simulation for electrically induced instability of a $160 \times 4$ DE film. (a) Undeformed film (b) Compressed film (in this case $\varepsilon^{pre} = 0.2$) (c) Wrinkled structure due to applied electric field and surface tension (here $\gamma/(\mu H_f) = 8$).

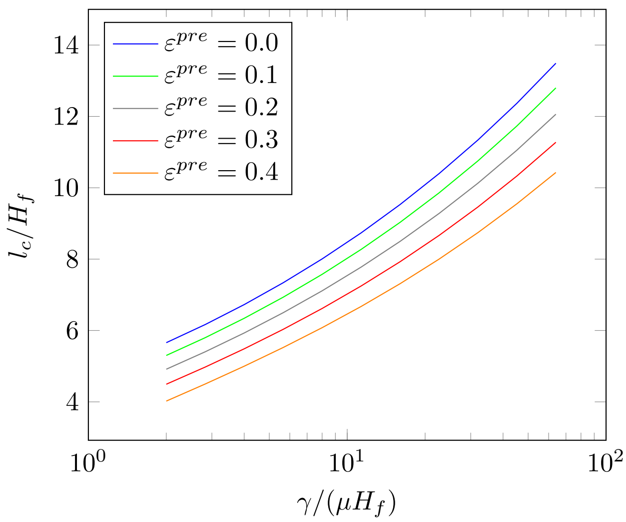



Figure 9: Analytic solution of critical wavelength vs. elasto-capillary number.

Incorporating reference-frame tracking and preserving the background hydrostatic pre-stress allows our formulation to achieve comprehensive physical symmetry, serving as a reliable analytical foundation for advanced active soft-matter designs.

## 5. Acknowledgements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